higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

POWER MACHINES N6

2 DECEMBER 2019

This marking guideline consists of 8 pages.

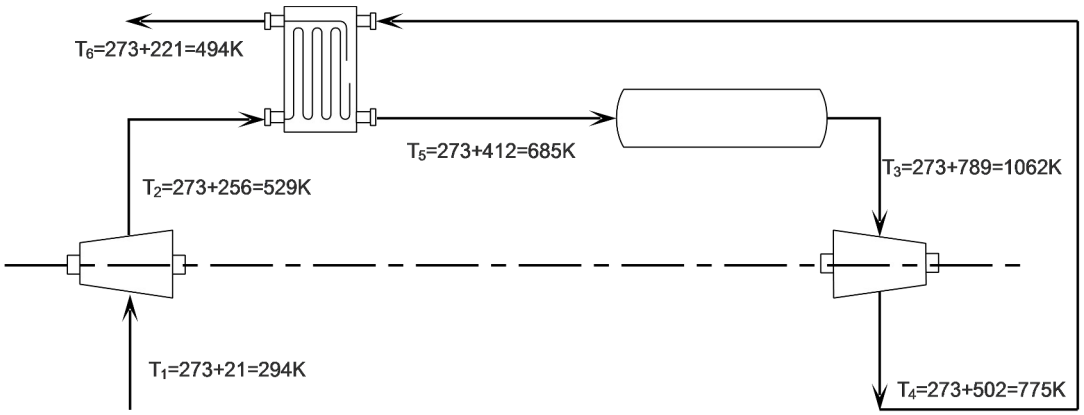
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✓ = 1 mark

✓½ = ½ mark

QUESTION 1

1.1



$$P_c = mC_p\left(T_2 - T_1\right) \quad ✓$$
$$= 7 \times 1{,}005\left(529 - 294\right) \quad ✓$$
$$= 1\,653{,}225kW \quad ✓$$

(3)

1.2
$$P_t = mC_p\left(T_4 - T_3\right) \quad ✓$$
$$= 7 \times 1{,}005\left(775 - 1\,062\right) \quad ✓$$
$$= -2\,019{,}045kW \quad ✓$$

(3)

1.3
$$P_{cc} = mC_p\left(T_3 - T_5\right) \quad ✓$$
$$= 7 \times 1{,}005\left(1\,062 - 685\right) \quad ✓$$
$$= 2\,652{,}195kW \quad ✓$$

(3)

1.4
$$\eta_{th.} = \frac{P_t - P_c}{P_{cc}} \times 100\%$$
$$= \frac{2\,019{,}045 - 1\,653{,}225}{2\,652{,}195} \times 100 \quad ✓$$
$$= 13{,}793\% \quad ✓$$

OR

$$\eta_{th.} = \frac{T_3 - T_4 - T_2 + T_1}{T_3 - T_5} \times 100\%$$
$$= \frac{1\,062 - 775 - 529 + 294}{1\,062 - 685} \times 100\%$$
$$= 13{,}793\%$$

(2)

1.5
$$\eta_{H.E.} = \frac{T_5 - T_2}{T_4 - T_6} \times 100\% \quad ✓$$
$$= \frac{685 - 529}{775 - 221} \times 100\% \quad ✓$$
$$= 28{,}159\% \quad ✓$$

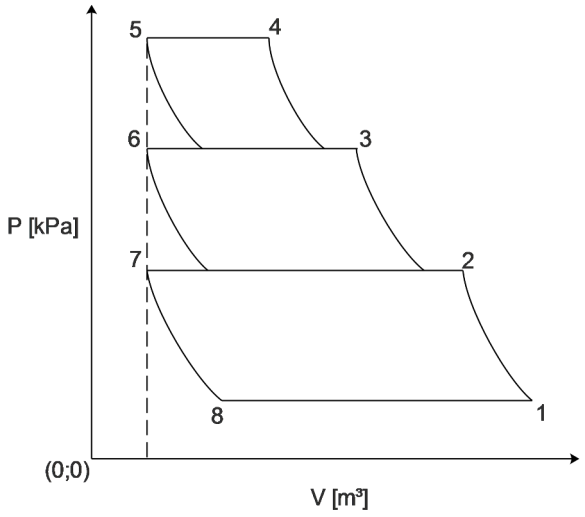
(3)

1.6
$$T_4' = T_3\left(\frac{P_4}{P_3}\right)^{\left(\frac{\gamma-1}{\gamma}\right)} \quad \checkmark$$
$$= 1\,062\left(\frac{1}{5}\right)^{\left(\frac{0,4}{1,4}\right)} \quad \checkmark$$
$$= 670,531K \quad \checkmark$$
(3)

1.7
$$\eta_t = \frac{T_3 - T_4}{T_3 - T_4'} \times 100\% \quad \checkmark$$
$$= \frac{1\,062 - 775}{1\,062 - 670,531} \times 100\% \quad \checkmark$$
$$= 73,314\% \quad \checkmark$$
(3)
[20]

QUESTION 2

2.1
$$P_1 V_{e1} \times N = mRT_1 \quad \checkmark$$
$$105 \times V_{e1} \times 300 = 16 \times 0,287 \times 294 \quad \checkmark$$
$$V_{e1} = 0,043 m^3 / cycle \quad \checkmark$$



(3)

2.2 2.2.1
$$P_2 V_{e2} \times N = mRT_2 \quad \checkmark$$
$$(105 \times 4) \times V_{e2} \times 300 = 16 \times 0,287 \times 294 \quad \checkmark$$
$$V_{e2} = 0,010\,7 m^3 / cycle \quad \checkmark$$
(3)

2.2.2
$$V_s = \frac{\pi}{4} D^2 L \quad \checkmark$$
$$= \frac{\pi}{4} \times 0,24^2 \times 0,296 \quad \checkmark$$
$$= 0,013\,4 m^3 / cycle \quad \checkmark$$
(3)

2.2.3
$$\eta_{vol2.} = \frac{V_{e2}}{V_{s2}} \times 100\%$$
$$= \frac{0,010\ 7}{0,013\ 4} \times 100\% \quad ✓$$
$$= 79,851\% \quad ✓$$
(2)

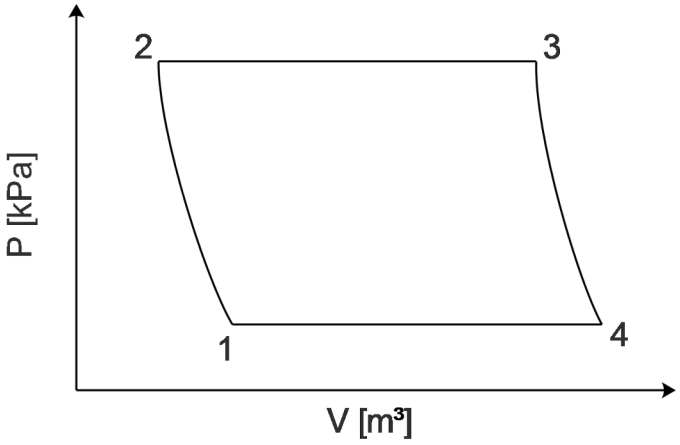
2.2.4
$$\eta_{vol.2} = 1 - \frac{V_{c2}}{V_{s2}}\left[\left(r_{ps}\right)^{\left(\frac{1}{n}\right)} - 1\right] \quad ✓$$
$$0,798\ 51 = 1 - \frac{V_{c2}}{0,013\ 4}\left[\left(4\right)^{\left(\frac{1}{1,32}\right)} - 1\right] \quad ✓$$
$$V_{c2} = 0,001\ 45 m^3 / cycle \quad ✓$$
(3)

2.2.5
$$V_2 = V_{c2} + V_{s2}$$
$$= 0,001\ 45 + 0,013\ 4 \quad ✓$$
$$= 0,014\ 85 m^3 / cycle \quad ✓$$
(2)

2.3
$$P = \frac{xn}{n-1}\dot{m}RT_1\left[\left(r_{ps}\right)^{\left(\frac{n-1}{n}\right)} - 1\right] \quad ✓$$
$$= \frac{3 \times 1,32}{0,32} \times \frac{16}{60} \times 0,287 \times 294\left[\left(4\right)^{\left(\frac{0,32}{1,32}\right)} - 1\right] \quad ✓✓$$
$$= 111,223 kW \quad ✓$$
(4)
[20]

QUESTION 3

3.1
$$T_1 = T_2\left(\frac{P_1}{P_2}\right)^{\left(\frac{\gamma-1}{\gamma}\right)} \quad ✓$$
$$= 471\left(\frac{110}{490}\right)^{\left(\frac{0,4}{1,4}\right)} \quad ✓$$
$$= 307,361 K \quad ✓$$



(3)

3.2
$$T_3 = T_1 + \Delta T$$
$$= 307,361 + 420 \quad ✓$$
$$= 727,361 K \quad ✓$$
(2)

3.3 $Q_{2-3} = mC_p(T_3 - T_2)$ ✓
$= 1 \times 1,005(727,361 - 471)$ ✓
$= 257,643kJ/kg$ ✓ (3)

3.4 $T_4 = T_3\left(\frac{P_4}{P_3}\right)^{\left(\frac{\gamma-1}{\gamma}\right)}$ ✓
$= 727,361\left(\frac{110}{490}\right)^{\left(\frac{0,4}{1,4}\right)}$ ✓
$= 474,655K$ ✓ (3)

3.5 $Q_{4-1} = mC_p(T_1 - T_4)$ ✓
$= 1 \times 1,005(307,361 - 474,655)$ ✓
$= -168,13kJ/kg$ ✓ (3)

3.6 $C_v = \frac{C_p}{\gamma}$
$= \frac{1,005}{1,4}$ ✓
$= 0,718kJ/kg.K$ ✓ (2)

3.7 $R = C_p - C_v$
$= 1,005 - 0,718$ ✓
$= 0,287kJ/kg.K$ ✓ (2)

3.8 $\eta = \frac{Q_{rec.} - Q_{rej.}}{Q_{rec}} \times 100\%$ $\eta = 1 - \frac{Q_{rej.}}{Q_{rec.}} \times 100\%$ $\eta = 1 - \left(\frac{P_1}{P_2}\right)^{\left(\frac{\gamma-1}{\gamma}\right)}$

$= \frac{257,643 - 168,13}{257,643} \times 100\%$ ✓ OR $\eta = 1 - \frac{168,13}{257,643} \times 100\%$ OR $= 1 - \left(\frac{110}{490}\right)^{\left(\frac{0,4}{1,4}\right)}$

$= 34,743\%$ ✓ $= 34,743\%$ $= 34,743\%$ (2)

[20]

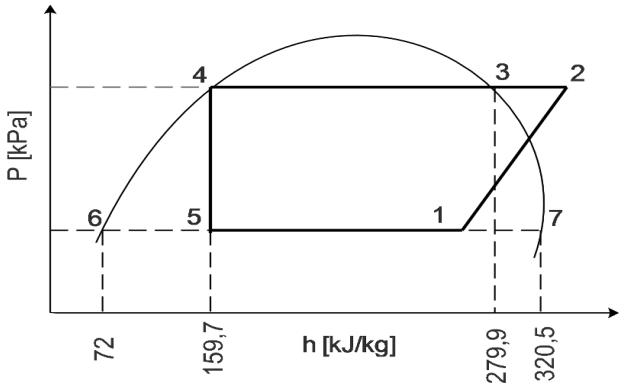
QUESTION 4

4.1

$$\eta_{vol.} = \frac{V_e}{V_s} \times 100\%$$

$$0,9 = \frac{V_e}{1\,500 \times 10^{-6}} \times 100\% \quad ✓$$

$$V_e = 0,001\,35 m^3 \quad ✓$$



(2)

4.2

$$V_e = \frac{V_g \times m_r \times x_1}{N} \quad ✓$$

$$0,001\,35 = \frac{0,012\,2 \times m_r \times 0,9}{122} \quad ✓$$

$$m_r = 15 kg / \min \quad ✓$$

(3)

4.3

$$P_c = \frac{m_r \times WD}{\eta_{mech.}} \quad ✓$$

$$(9,25 \times 60) = \frac{15 \times WD}{0,9} \quad ✓$$

$$WD = 29,6 kJ / \min \quad ✓$$

(3)

4.4

$$h_1 = h_6 + x_1(h_7 - h_6) \quad ✓$$

$$= 72 + 0,9(320,5 - 72) \quad ✓$$

$$= 295,65 kJ / kg \quad ✓$$

(3)

4.5

$$WD = h_2 - h_1$$

$$29,6 = h_2 - 295,65 \quad ✓$$

$$h_2 = 325,25 kJ / kg \quad ✓$$

(2)

4.6

$$h_2 = h_3 + C_p(T_2 - T_3) \quad ✓$$

$$325,25 = 279,9 + 1,814(T_2 - 298) \quad ✓$$

$$T_2 = 323,138 K \quad ✓$$

$$= 50,138°C \quad ✓$$

(4)

4.7

$$Q_{con.} = m_r(h_2 - h_4) \quad ✓$$

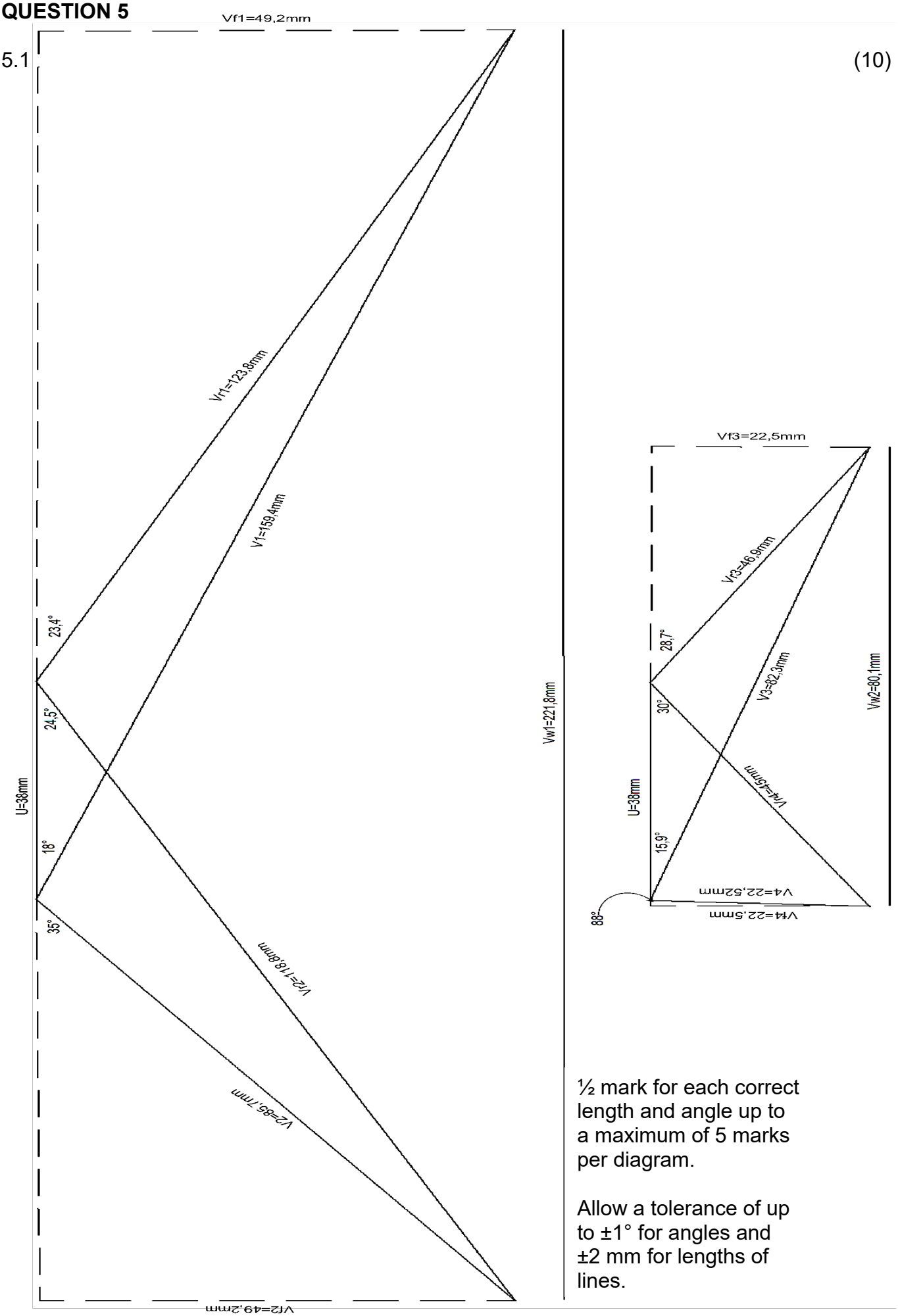
$$= 15(325,25 - 159,7) \quad ✓$$

$$= 2\,483,25 kJ / \min \quad ✓$$

(3)
[20]

QUESTION 5

5.1 (10)



½ mark for each correct length and angle up to a maximum of 5 marks per diagram.

Allow a tolerance of up to ±1° for angles and ±2 mm for lengths of lines.

5.2 5.2.1 $V_2 = 85,7 \times 4 = 342,8 m/s$ ✓ (1)

5.2.2 $V_3 = 82,3 \times 4 = 329,2 m/s$ ✓ (1)

5.2.3 $\theta_1 = 23,4°$ ✓ (1)

5.2.4 $\phi_1 = 24,5°$ ✓ (1)

5.2.5 $\theta_2 = 28,7°$ ✓ (1)

5.2.6 $\alpha_1 = 18°$ ✓ (1)

5.2.7 $\alpha_2 = 15,9°$✓ (1)

5.2.8 $P = mUV_{WT}$
$= 1 \times 152(221.8 + 80.1) \times 4$ ✓
$= 183\ 555,2 J/kg$ ✓
$183,556 kJ/kg$ ✓ (3)
[20]

TOTAL: 100

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